

WIRSCHING’S CONJECTURE 3, AND THE PERIODIC CORRECTION TO BERG–KRÜPPEL’S DENSITY ASYMPTOTIC

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ABSTRACT. Wirsching’s 2003 argument for positive predecessor density under the $3n + 1$ map reduces to three conjectures. Conjecture 3 concerns the near-zero asymptotics of an invariant density φ , compared against an explicit asymptotic φ_0 due to Berg and Krüppel. The exact log-Laplace transform of φ decomposes as a smooth part plus a periodic correction H . Berg and Krüppel gave this correction, for this eigenfunction, as an infinite product in 1998, five years before Wirsching’s conjecture; they did not ask whether it is constant. We give H instead as a Fourier series with closed-form coefficients in Γ and ζ , and certify rigorously that it is not. Wirsching’s own comparison class fixes a single phase of H ; there, Conjecture 3 holds, with an explicit limit. Away from that class the phase sweeps a full period, so the stronger, unrestricted asymptotic $\varphi(t) \sim c \varphi_0(t)$ as $t \rightarrow 0^+$ fails. Wirsching needs less than the literal statement of Conjecture 3; we check directly that his actual requirement holds with a wide margin.

1. INTRODUCTION

Let T denote the $3n+1$ map on the positive integers, $T(n) = n/2$ for n even and $T(n) = (3n+1)/2$ for n odd. Wirsching’s monograph [3] develops the predecessor-set machinery this paper’s argument descends from; [1] takes it further, studying the density of predecessor sets under T through an averaging construction on the 3-adic integers: a family of operators S_l , built from path counts in the predecessor graph, converges in the strong operator topology to a limiting operator S_∞ whose essential part is a Markov chain on $[0, 1]$. That chain has a unique invariant density φ , and Wirsching’s proof of positive predecessor density reduces to three conjectures about φ and related quantities, stated as three unbridged gaps in the argument.

The load-bearing one is Conjecture 3. It concerns the behaviour of φ as its argument tends to 0, compared against an explicit comparison function φ_0 . Berg and Krüppel [2] had already produced φ_0 five years earlier, by Laplace transform and saddlepoint analysis, as the asymptotic solution of a truncated version of the functional equation φ satisfies. Their argument for the asymptotic comparability of φ and φ_0 was informal, and left open both the exact multiplicative constant and whether the ratio converges at all. Wirsching considers the strongest possible reading of that gap directly:

“That replacement would be possible, if, e.g., $\varphi \sim c \varphi_0$ for some constant $c > 0$. It turns out that the following (slightly weaker) conjecture . . . suffices.”

He then states Conjecture 3: along a specific comparison class of sequences (z_l) tending to 0 at the borderline rate $z_l \sim l \cdot 3^{-l}$, the ratio $\varphi(z_l)/\varphi_0(z_l)$ converges to a positive constant, uniformly on the class. This is weaker than the unrestricted asymptotic he first considers. The constant cancels out of the one inequality his own proof actually needs, so a class-dependent limit is enough.

Neither statement had been settled. This paper settles both, in opposite directions.

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Theorem 1 (Conjecture 3). *Wirsching's Conjecture 3 is true. On his comparison class $\tilde{A}_\delta$, the ratio $\varphi(z_l)/\varphi_0(z_l)$ converges uniformly to $e^{H(0)} = 0.534122\dots$, where H is the periodic function of Theorem 4 below and the normalization of φ_0 is Berg and Krüppel's own, equation (9.6) of [2].*

Theorem 2 (Failure of the unrestricted asymptotic). *The stronger asymptotic $\varphi(t) \sim \kappa \varphi_0(t)$ as $t \rightarrow 0^+$, for any single constant κ and with no restriction to a comparison class, is false. The ratio $\varphi(t)/\varphi_0(t)$ oscillates indefinitely as $t \rightarrow 0^+$, with period $\log 3$ in $\log(1/t)$ and amplitude bounded below by a positive, certified constant.*

The two theorems have a single common source. Write $t = e^{-\tau}$ and let $H(w)$ be the $\log 3$ -periodic function appearing in the exact decomposition of $\log \varphi$'s log-Laplace transform (Theorem 4). Wirsching's comparison class fixes τ 's phase modulo $\log 3$ at a single value; every other class of comparison sequences sweeps through all phases. Whether the ratio converges or oscillates depends on which sequences one compares along, and the two theorems above read the same object at two different scopes.

Corollary 3. *H is Berg and Krüppel's own periodic factor for the eigenfunction φ , from Proposition 9.3 of [2], which they give as an infinite product. Its Fourier series, Proposition 6 below, is new.*

Section 2 recalls the functional equation φ satisfies and Berg and Krüppel's comparison asymptotic. Section 3 constructs H , proves Proposition 6 and Corollary 3, and certifies that H is not constant. Sections 4 and 5 transfer this from the log-Laplace transform to φ itself, through a uniform saddlepoint approximation and an envelope estimate. Section 6 identifies the resulting smooth part with Berg and Krüppel's saddlepoint expression and pins down every constant. Section 7 proves both theorems and checks Wirsching's actual requirement directly. Section 8 states what remains open.

2. THE INVARIANT DENSITY AND ITS LAPLACE TRANSFORM

Let $U_1, U_2, \dots$ be independent, identically distributed uniform random variables on $[0, 1]$, and set

$$X = \sum_{j \geq 1} \frac{2}{3^j} U_j \in [0, 1].$$

The self-similar identity $X = \frac{2}{3}U_1 + \frac{1}{3}X'$, with X' an independent copy of X , shows that the density φ of X satisfies

$$\varphi(x) = \frac{3}{2} \int_{3x-2}^{3x} \varphi,$$

equivalently $\varphi'(x) = \frac{9}{2}\varphi(3x)$ on $(0, \frac{2}{3})$: this is Wirsching's invariant density (up to the reflection $x \mapsto 1-x$ fixing the convention), a translate of Rvachev's atomic function h_3 [4], and the dilation-3, $\lambda = 2/3$ member of Berg and Krüppel's family of functional equations $\lambda g'(t) = 3(g(3t) - g(3t-2))$. It is C^∞ , supported on $[0, 1]$, and constant, equal to $3/2$, on the middle third $[1/3, 2/3]$.

For $s > 0$ write

$$K(s) = \log \mathbb{E}[e^{-sX}] = \sum_{j \geq 1} g\left(\frac{2s}{3^j}\right), \quad g(b) = \log \frac{1 - e^{-b}}{b}.$$

The branch of g is the one analytic on $\{\operatorname{Re} w > 0\}$ and real on $(0, \infty)$; since $(1 - e^{-w})/w$ is analytic and non-vanishing there (its zeros lie on $2\pi i\mathbb{Z} \setminus \{0\}$, and $w = 0$ is a removable singularity with value 1), and the half-plane is simply connected, g exists and is unique. Differentiating termwise, justified because $g(w) = -w/2 + O(w^2)$ as $w \rightarrow 0$ so the tail of the series is normally convergent together with every derivative,

$$t(s) := -K'(s), \quad V(s) := s^2 K''(s) > 0,$$

the second an exact positivity, $K''(s)$ being the variance of X under its exponential tilt by e^{-sX} . As s ranges over $(0, \infty)$, $t(s)$ decreases strictly from $1/2$ to 0 (proved in Lemma 15 below), so studying $\varphi(t(s))$ as $s \rightarrow \infty$ is the same as studying $\varphi(t)$ as $t \rightarrow 0^+$.

Berg and Krüppel's comparison function solves the truncated equation $\lambda g'(t) = dg(dt)$ by Laplace transform and saddlepoint analysis. In their normalization (their equation (9.6), with $\alpha = \frac{1}{2} - \frac{\log 2}{\log 3}$, $\beta = \frac{1}{2\log 3}$, and $\gamma, \delta_{\text{BK}}, \varepsilon$ determined by α and β through their Proposition 9.1),

$$\varphi_0(t) = \frac{(2\beta)^\varepsilon}{\sqrt{2\pi}} t^\gamma (-\log t)^{\delta_{\text{BK}}} \exp\left(-\beta \log^2 \frac{t}{-\log t}\right).$$

Write $\varphi_{0,\text{bare}}$ for the same expression without the leading constant $(2\beta)^\varepsilon/\sqrt{2\pi}$: the two differ only by that constant.

3. THE PERIODIC CORRECTION

Set $c = \log 3$ and write $w = \log s$, $L(w) = K(e^w)$. Substituting into the defining series and splitting off the $j = 1$ term against the rest reproduces the recurrence

$$(1) \quad L(w+c) - L(w) = \log(1 - e^{-2e^w}) - w - \log 2.$$

Let

$$Q(w) = -\frac{w^2}{2c} + \left(\frac{1}{2} - \frac{\log 2}{c}\right)w,$$

chosen so that $Q(w+c) - Q(w) = -w - \log 2$, matching (1)'s non-periodic terms exactly. Write $a := \frac{1}{2} - \frac{\log 2}{c}$ for Q 's linear coefficient, used again from Section 5 onward.

Theorem 4. *Define*

$$(2) \quad H(w) := L(w) - Q(w) + \sum_{k \geq 0} \log(1 - e^{-2 \cdot 3^k e^w}).$$

Then H is exactly c -periodic, $H(w+c) = H(w)$ for every $w \in \mathbb{R}$, and

$$L(w) = Q(w) + H(w) + \Delta(w), \quad \Delta(w) = -\sum_{k \geq 0} \log(1 - e^{-2 \cdot 3^k e^w}),$$

with $\Delta^{(j)}(w) = O_j(e^{jw-2e^w})$ for every $j \geq 0$: doubly exponentially small, together with all its derivatives, as $w \rightarrow +\infty$.

Proof. Write $d(w) = \log(1 - e^{-2e^w})$, so (1) reads $L(w+c) - L(w) = d(w) - w - \log 2$. Directly from the definition of Q (expand $(w+c)^2 = w^2 + 2wc + c^2$ and collect terms), $Q(w+c) - Q(w) = -w - \log 2$ as well, so the two non-periodic parts cancel exactly:

$$(L - Q)(w+c) - (L - Q)(w) = [d(w) - w - \log 2] - [-w - \log 2] = d(w).$$

The series in (2) telescopes this exactly: with $D(w) := (L - Q)(w)$,

$$D(w) + \sum_{k \geq 0} d(w + kc) = D(w+c) - d(w) + \sum_{k \geq 0} d(w + kc) = D(w+c) + \sum_{k \geq 0} d(w + (k+1)c),$$

so the right side of (2) is invariant under $w \mapsto w+c$, that is, H is c -periodic. Convergence of the series and its termwise differentiability, together with the stated bound on $\Delta^{(j)}$, follow from $d(w) = O(e^{-2e^w})$ and the corresponding bounds on its derivatives, since $2 \cdot 3^k e^w \geq 2e^w$ for $k \geq 0$ makes the series dominated, uniformly for w in any half-line bounded away from $-\infty$, by a geometric series in 3^{-k} inside the exponent. $\square$

The decomposition $L = Q + H + \Delta$, with Δ negligible as $w \rightarrow \infty$, is unique: any two periodic functions differing from $L - Q$ by an $o(1)$ term as $w \rightarrow \infty$ must be equal, being periodic functions that agree in the limit along every residue class of w modulo c .

Remark 5. Berg and Krüppel's own Proposition 9.3 (specialized to $a = 3$, $b = 3/2$ in their notation, exactly this eigenfunction) gives the equivalent identity $\exp(H(w)) = a^{\frac{1}{2}t^2 - \alpha t} \prod_{k \geq 0} \frac{1 - e^{-a^{t-k}/b}}{a^{t-k}/b} \prod_{l \geq 1} (1 - e^{-a^{t+l}/b})$, $t = w/c$, as an infinite product, in their proof of that proposition. The Fourier series of Proposition 6 below is what lets us certify non-constancy (Proposition 8), which their product form does not make transparent.

Proposition 6. *H has the Fourier expansion $H(w) = \sum_{m \in \mathbb{Z}} \hat{H}(m) e^{i\omega_m w}$, $\omega_m = 2\pi m/c$, with*

$$\hat{H}(0) = \frac{\log 2}{2} - \frac{\log^2 2}{2c} - \frac{c}{12} - \frac{A}{c}, \quad A = \frac{\pi^2}{12} - \frac{\gamma_E^2}{2} - \gamma_1,$$

$$\hat{H}(m) = -\frac{2^{i\omega_m}}{c} \Gamma(-i\omega_m) \zeta(1 - i\omega_m), \quad m \neq 0,$$

where γ_E is the Euler–Mascheroni constant and γ_1 the first Stieltjes constant.

Sketch¹. The Mellin transform of g on the strip $-1 < \operatorname{Re} z < 0$ is $g^*(z) = -\Gamma(z)\zeta(1+z)$, obtained by integrating by parts onto $g'(b) = 1/(e^b - 1) - 1/b$ and using the classical continuation $\int_0^\infty (\frac{1}{e^u - 1} - \frac{1}{u}) u^{z-1} du = \Gamma(z)\zeta(z)$. Since $K(s) = \sum_j g(2s/3^j)$, the Mellin transform of $s \mapsto K(s)$ picks up a factor $2^z \sum_j 3^{-jz} = 2^z/(3^z - 1)$ from this harmonic sum; shifting the resulting inversion contour past the triple pole at the origin and past each simple pole at $z = i\omega_m$ (where $3^z = 1$) reproduces $Q + \hat{H}(0)$ and the Fourier modes above, respectively, the factor $2^{i\omega_m}$ coming from that same harmonic-sum step. $\square$

Remark 7. This is an instance of the de Bruijn–Mahler phenomenon: a base-change harmonic sum, Mellin transformed, picks up simple poles on a vertical line, and their residues assemble into an exactly periodic correction with Fourier coefficients in Γ and ζ . De Bruijn found the first case of this in Mahler's partition asymptotic [5]; Erdős and Richmond named it [6]; Kato and McLeod [7] and Derfel [8] established the log-squared decay rate of (2)'s companion asymptotic φ_0 as generic across this whole class of rescaled functional equations. Our contribution here is the closed form for the periodic correction itself, not the mechanism producing it.

The series for H' and H'' converge absolutely and uniformly, since $|\Gamma(-i\omega_m)| = \sqrt{\pi/(\omega_m \sinh \pi\omega_m)}$ decays exponentially in m while $\zeta(1 - i\omega_m)$ grows only polynomially, and this yields explicit, certified bounds:

$$(3) \quad \sup_w |H'(w)| \leq 0.0011977472315550332, \quad \sup_w |H''(w)| \leq 0.0068518962896650951.$$

Proposition 8 (Certified non-constancy). *H is not constant. Rigorously,*

$$-0.000377190280943987 < H(0) - H(\log(3/2)) < -0.000377190280943985,$$

and consequently $\operatorname{osc}(H) := \sup H - \inf H \geq 3.7719 \times 10^{-4}$; a full optimization gives the certified enclosure $\operatorname{osc}(H) \in [4.18744947692 \times 10^{-4}, 4.18756644224 \times 10^{-4}]$.

Sketch. Write $g(b) = -b + \log S(b)$, $S(b) = \sinh(b)/b = \sum_{n \geq 0} b^{2n}/(2n+1)!$. Termwise domination of $6^n n! \leq (2n+1)!$ gives $0 \leq \log S(b) \leq b^2/6$ for $b \geq 0$, hence an explicit, two-sided enclosure of the tail $\sum_{r > R} g(s/3^r)$ for any R and $s > 0$; combined with a finite head evaluated in interval or ball arithmetic (via (2) directly, or independently via ball arithmetic in Γ and ζ applied to Proposition 6), this certifies the stated interval to arbitrary precision. The full oscillation is certified the same way, via the Lipschitz bound in (3) and a fine grid search over one period. $\square$

4. A UNIFORM SADDLEPOINT APPROXIMATION

Define, for $s > 0$,

$$\varphi_{\text{sp}}(t(s)) := \frac{\exp(K(s) + st(s))}{\sqrt{2\pi K''(s)}}.$$

This section proves that $\varphi(t(s))/\varphi_{\text{sp}}(t(s)) \rightarrow 1$ as $s \rightarrow \infty$, uniformly over every phase of s modulo $3\mathbb{Z}$, which is what makes the two theorems of Section 7 depend on different scopes rather than on different mathematics.

For $n \geq 2$, define the derivative-normalized functions $\kappa_n(w) = (-1)^n w^n g^{(n)}(w)$ on $\{\text{Re } w > 0\}$; $\kappa_n(b)$, $b > 0$, is b^n times the n -th cumulant of the random variable with density $b e^{-bw}/(1 - e^{-b})$ on $[0, 1]$. Writing $x = w/2$,

$$\kappa_2(w) = 1 - \frac{x^2}{\sinh^2 x}, \quad \kappa_3(w) = 2 - \frac{2x^3 \cosh x}{\sinh^3 x}, \quad \kappa_4(w) = 6 + \frac{2x^4(1 - 3 \coth^2 x)}{\sinh^2 x},$$

and from the Bernoulli expansion $g(w) + w/2 = \sum_{k \geq 1} B_{2k} w^{2k}/(2k(2k)!)$ on $|w| < 2\pi$,

$$\kappa_n(w) = (-1)^n \sum_{k \geq \lceil n/2 \rceil} \frac{B_{2k} w^{2k}}{2k(2k-n)!}, \quad n \geq 2,$$

so that $\kappa_n(w) = O(w^2)$ as $w \rightarrow 0$ for every $n \geq 2$.

Write $s = \rho \cdot 3^N/2$ for the unique $N = \lfloor \log_3(2s) \rfloor \in \mathbb{Z}$ and $\rho \in [1, 3)$, and $b_j = 2s/3^j$. Then $\{b_j : j \geq 1\} = \{\rho \cdot 3^m : m \leq N-1\}$: exactly N of these, the smallest being $\rho \geq 1$, are ≥ 1 ; the rest form a geometric tail below 1. This single structural fact is the source of every phase-independent constant below.

Lemma 9 (Boundedness off the real axis). *For $w = b(1+iu)$, $b > 0$, $|u| \leq 1$, with $e(r) = r^2/\sinh^2 r$ and $f(r) = r^3 \cosh r/\sinh^3 r$,*

$$|\kappa_2(w)| \leq 1 + 2e(b/2), \quad |\kappa_3(w)| \leq 2 + 2 \cdot 2^{3/2} f(b/2).$$

Both bounds are finite over the whole sector $|\text{Im } w| \leq \text{Re } w$; $M_2 := \sup |\kappa_2| \leq 3$, $M_3 := \sup |\kappa_3| \leq 2 + 2^{5/2} = 7.657$. In addition, for $|w| \leq 2$, $|\kappa_2(w)| \leq 0.114|w|^2$ and $|\kappa_3(w)| \leq 0.0119|w|^4$.

Proof. $|\sinh z|^2 = \sinh^2(\text{Re } z) + \sin^2(\text{Im } z) \geq \sinh^2(\text{Re } z)$ and $|\cosh z|^2 \leq \cosh^2(\text{Re } z)$; the first excludes the poles of $\coth$, since $\text{Re}(w/2) = b/2 > 0$ never approaches $\pi i k$. This gives the sector bounds directly. Monotonicity of e is immediate from the increasing power series of $\sinh(r)/r$; monotonicity of f is Lazarević's inequality, $(\sinh r/r)^3 > \cosh r$, checked by comparing alternating truncated series near $r = 0$ and by monotonicity of $\coth r - 1/r$ away from it. The local bounds at $|w| \leq 2$ follow by applying the maximum-modulus principle to $\kappa_2(w)/w^2$ and $\kappa_3(w)/w^4$ (analytic on $|w| < 2\pi$ by the Bernoulli expansion): bound each on $|w| = 2$ by the triangle inequality, using $|B_{2k}| = 2(2k)! \zeta(2k)/(2\pi)^{2k}$ and $\zeta(2k) \leq \zeta(2)$, respectively $\zeta(4)$. $\square$

Lemma 10 (Variance). *For all $\rho \in [1, 3)$, $N \geq 1$, with $V := s^2 K''(s) = \sum_j \kappa_2(b_j)$,*

$$N - A_V \leq V \leq N + 0.1283, \quad A_V := \sum_{m \geq 0} e(3^m/2) = 1.4269413069 \dots$$

Proof. Split $V = \sum_{m=0}^{N-1} \kappa_2(\rho 3^m) + \sum_{m < 0} \kappa_2(\rho 3^m)$, using $\kappa_2(b) = 1 - e(b/2) \in (0, 1)$. For the lower bound, drop the second (non-negative) sum and use $e(\rho 3^m/2) \leq e(3^m/2)$ (e decreasing, $\rho \geq 1$) in the first, giving $\sum_{m=0}^{N-1} \kappa_2(\rho 3^m) \geq N - A_V$. For the upper bound, the first sum is $< N$ term by term, and the second sum, where $\rho 3^m < 1$, is bounded by Lemma 9's local estimate: $\sum_{m < 0} \kappa_2(\rho 3^m) \leq 0.114\rho^2 \sum_{m < 0} 9^m = 0.114\rho^2/8 \leq 0.1283$ for $\rho < 3$. $\square$

Lemma 11 (Tail bound and existence of the density). *If $b_0 \geq 1$ and $b_k = 3^k b_0$, then $\sum_{k \geq 0} \log \coth(b_k/2) \leq 3e^{-b_0}$. Consequently, with $\Lambda(y) := K(s(1+iy)) - K(s) + isyt(s)$,*

$$|e^{\Lambda(y)}| \leq e^{3/e} (1+y^2)^{-N/2} \leq 3.0152 (1+y^2)^{-N/2}$$

for every real y . In particular X has a continuous density, the tilted characteristic function is integrable for $N \geq 3$, and Fourier inversion at $x = t(s)$ gives

$$(4) \quad R(s) := \frac{\varphi(t(s))}{\varphi_{\text{sp}}(t(s))} = \sqrt{\frac{V}{2\pi}} \int_{\mathbb{R}} e^{\Lambda(y)} dy,$$

a real quantity, since $\Lambda(-y) = \overline{\Lambda(y)}$.

Proof. $\log \coth(x/2) = 2 \operatorname{artanh}(e^{-x}) \leq 2e^{-x}/(1 - e^{-2x})$; summing over the geometric orbit $b_k = 3^k b_0$ and bounding the resulting series by its first few terms gives the stated constant. X 's density is continuous because the sum of the first two summands already has a continuous, compactly supported density and convolution preserves continuity. For Λ : $t(s) = -K'(s)$ cancels the linear term of the tilted transform exactly, and each remaining factor $e^{g(b_j(1+iy)) - g(b_j)}$ has modulus at most 1 (a tilted characteristic function), with modulus at most $\coth(b_j/2)/\sqrt{1+y^2}$ on the N indices where $b_j \geq 1$; the tail estimate applies to exactly that geometric set, with $b_0 = \rho \geq 1$. $\square$

Lemma 12 (Central expansion). *With $h(u) = g(b(1+iu))$, Taylor's theorem in integral form (the Lagrange form is invalid for a complex-valued function of a real variable) gives*

$$h(y) = h(0) + h'(0)y + h''(0)\frac{y^2}{2} + \frac{1}{2} \int_0^y (y-u)^2 h'''(u) du.$$

Consequently, for $\theta \in (0, 1]$,

$$S := \sum_j \sup_{|u| \leq \theta} |h_j'''(u)| \leq 2N + 3.7442,$$

and hence $|\Lambda(y) + Vy^2/2| \leq B|y|^3$ for $|y| \leq \theta$, with $B := S/6 \leq (2N + 3.7442)/6$.

Proof. Write $h_j(u) := g(b_j(1+iu))$ for each j , so $h_j'''(u) = \kappa_3(b_j(1+iu))/(1+iu)^3$ up to a sign. The bound on h_j''' follows from Lemma 9 applied to κ_3 , summed over the N indices with $b_j \geq 1$ (via f 's monotonicity, worst case $\rho = 1$) and the remaining geometric tail (via the local bound). $\square$

Theorem 13 (Formula (A)). *For $N \geq 17$, uniformly over $\rho \in [1, 3]$, $|R(s) - 1| \leq E(N) := e_1(N) + e_2(N) + e_3(N)$, with $\theta_N := N^{-1/3}$, $V_{\text{lo}} := N - A_V$, $V_{\text{up}} := N + 0.1283$, $B := (2N + 3.7442)/6$,*

$$(5) \quad \begin{aligned} e_1(N) &:= \frac{4e^{B\theta_N^3} B}{\sqrt{2\pi} V_{\text{up}}^{3/2}}, & e_2(N) &:= \frac{2}{\sqrt{2\pi}} \cdot \frac{e^{-\theta_N^2 V_{\text{lo}}/2}}{\theta_N \sqrt{V_{\text{lo}}}}, \\ e_3(N) &:= 2e^{3/e} \sqrt{\frac{V_{\text{up}}}{2\pi}} \cdot \frac{(1 + \theta_N^2)^{-(N-2)/2}}{\theta_N(N-2)}. \end{aligned}$$

E is strictly decreasing, $E(17) < 1$, $E(N) \leq 4.1N^{-1/2}$ for all $N \geq 17$, and $\sqrt{N}E(N) \rightarrow \frac{4}{3}e^{1/3}(2\pi)^{-1/2} = 0.742358\dots$. Consequently $\varphi(t) = \varphi_{\text{sp}}(t)(1 + o(1))$ as $t \rightarrow 0^+$, with no restriction on the sequence of t 's.

Proof. Split (4) at $|y| = \theta_N$. On $|y| \leq \theta_N$, Lemma 12 bounds the cubic remainder $|\Lambda(y) + Vy^2/2|$ by $B\theta_N^3$, which stays bounded (not $o(1)$): this is exactly why e_1 carries the factor $e^{B\theta_N^3}$ rather than vanishing) as $N \rightarrow \infty$, since $B = O(N)$ and $\theta_N^3 = N^{-1}$. Writing $C(y) := e^{\Lambda(y)} - e^{-Vy^2/2}$, $|e^z - 1| \leq |z|e^{|z|}$ gives $|C(y)| \leq B|y|^3 e^{B\theta_N^3}$ on this range, and $\int_{\mathbb{R}} e^{-Vy^2/2} |y|^3 dy = 4/V^2$ (using $V \geq V_{\text{lo}}$, from Lemma 10) gives the central Taylor-remainder contribution e_1 . The truncated Gaussian tail $2 \int_{\theta_N}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-Vy^2/2} dy$, bounded via $\int_c^{\infty} e^{-v^2/2} dv \leq e^{-c^2/2}/c$ at $c = \theta_N \sqrt{V_{\text{lo}}}$, gives e_2 .

On $|y| > \theta_N$, Lemma 11's bound $|e^{\Lambda(y)}| \leq e^{3/e}(1+y^2)^{-N/2}$ integrates, via $\int_a^\infty (1+y^2)^{-N/2} dy \leq (1+a^2)^{-(N-2)/2}/(a(N-2))$ at $a = \theta_N$ (using $V \leq V_{\text{up}}$ for the leading $\sqrt{V/2\pi}$ factor in (4)), to give e_3 ; this needs $\theta_N \gg N^{-1/2}$, satisfied since $\frac{1}{3} < \frac{1}{2}$. Every constant entering E (via A_V , Lemma 10, and the $3.0152 \approx e^{3/e}$ of Lemma 11) is a supremum over $b > 0$ or a sum over one complete geometric orbit bounded using $\rho \geq 1$, hence independent of ρ . Evaluating (5) numerically gives the stated $N_0 = 17$, monotonicity, and limit. Since $t(s)$ ranges bijectively over $(0, 1/2)$, this is exactly the statement $\varphi(t) = \varphi_{\text{sp}}(t)(1 + o(1))$ as $t \rightarrow 0^+$. $\square$

Remark 14. The rate is not sharp. Carrying the central expansion to fourth order, using that the odd cubic term integrates to zero against the Gaussian core, and using the limits $V_3 := \sum_j \kappa_3(b_j) = 2N + O(1)$, $V_4 := \sum_j \kappa_4(b_j) = 6N + O(1)$ (the third and fourth cumulant sums, by the same argument as Lemma 10) gives the Edgeworth correction $R(s) = 1 - \frac{1}{12N} + O(N^{-3/2})$, matching the classical Stirling correction for a $\text{Gamma}(N, 1)$ law: after tilting, the $b_j \geq 1$ summands behave asymptotically like independent $\text{Exp}(1)$ variables, so X under its tilt is asymptotically $\text{Gamma}(N, 1)$ -distributed, and ρ enters only through $O(1)$ shifts of the cumulants, at order N^{-2} . We do not need this sharper rate and do not prove the remainder term here; Theorem 13's qualitative $o(1)$ is all that Section 7 uses.

5. THE ENVELOPE LEMMA

Theorem 13 compares φ to φ_{sp} , a quantity built from the full transform $K = Q + H + \Delta$. It remains to compare φ_{sp} to a smooth auxiliary system built from Q alone, so that the periodic part H can be extracted explicitly. Write $t = e^{-\tau}$, $g_\tau(w) = L(w) + e^{w-\tau}$, $g_{0,\tau}(w) = Q(w) + e^{w-\tau}$. The true and smooth systems each have their own analogue of $-K'$: write $B_{\text{tr}}(w) := e^{wt}(e^w) = -L'(w)$ and $B_{\text{sm}}(w) := -Q'(w) = w/c - a$.

Lemma 15 (True and smooth saddles). *For each τ , g_τ has a unique global minimizer $w^*(\tau)$, at which $g_\tau''(w^*) = V(r^*) > 0$, $r^* = e^{w^*}$; and $g_{0,\tau}$ has a unique local minimizer $w_0(\tau)$ among solutions with $B_0 := B_{\text{sm}}(w_0) = e^{w_0-\tau} > 1/c$. Both are characterized by $\tau = \Phi(w^*) = \Phi_0(w_0)$ for $\Phi(w) = w - \log B_{\text{tr}}(w)$ (a strictly increasing bijection $\mathbb{R} \rightarrow (\log 2, \infty)$, since $t(s)$ is strictly decreasing) and $\Phi_0(w) = w - \log B_{\text{sm}}(w)$, and satisfy*

$$|w^* - w_0| = O(1/B_0), \quad 0 \leq g_\tau(w_0) - g_\tau(w^*) = O(1/B_0).$$

Sketch. Strict convexity of K gives the global uniqueness of w^* . At w_0 , $g_\tau'(w_0) = H'(w_0) + O(e^{-2e^{w_0}})$ by Theorem 4, bounded by (3); since g_τ'' is bounded above and below by explicit constants on $[w_0 - 1, w_0 + 1]$ (again via (3) and Lemma 9), a sign-change argument for g_τ' locates w^* within $O(1/B_0)$ of w_0 , and Taylor's theorem about w^* then bounds the resulting gap in g_τ -values by the same order. $\square$

Proposition 16 (Envelope estimate). *With $S(\tau) := \log \varphi_{\text{sp}}(t) = g_\tau(w^*) + w^* - \frac{1}{2} \log(2\pi V(r^*))$ and $P(\tau) := Q(w_0) + B_0 + w_0 - \frac{1}{2} \log(2\pi(B_0 - \frac{1}{c}))$,*

$$|S(\tau) - P(\tau) - H(w_0(\tau))| = O(1/\tau) \quad (\tau \rightarrow \infty),$$

uniformly in the phase of τ modulo c .

Sketch. Expanding $S(\tau) - P(\tau) - H(w_0)$ exactly using $L = Q + H + \Delta$ and Lemma 15 produces four terms: the true envelope gap $g_{0,\tau}(w^*) - g_{0,\tau}(w_0)$, which is $O(B_0(w^* - w_0)^2) = O(1/B_0)$; the periodic-part difference $H(w^*) - H(w_0) = O(\eta|w^* - w_0|) = O(1/B_0)$ using (3); the saddle-location shift $w^* - w_0$ itself, $O(1/B_0)$; and a normalization term comparing $V(r^*)$ to $B_0 - 1/c$, again $O(1/B_0)$ using (3)'s bound on H'' . Since $\tau = w_0 - \log B_0$, $B_0/\tau \rightarrow 1/c$, and all four terms are $O(1/\tau)$ with constants free of the phase of τ . $\square$

6. THE BERG–KRÜPPEL IDENTITY

Proposition 17. *$P(\tau)$ is identical, as a function of τ , to the exact saddlepoint expression Berg and Krüppel construct in the proof of their Proposition 9.1 – the pre-asymptotic quantity their own argument shows is asymptotic, as $t \rightarrow 0^+$, to the explicit closed form φ_0 – not merely asymptotic to it. With $Y = cB_0$, $r = -ca - \log c$, the saddle equation for that construction is exactly $Y - \log Y = \tau + r$, and $P(\tau) = Q(w_0) + B_0 + w_0 - \frac{1}{2} \log(2\pi(B_0 - 1/c))$ coincides term for term with theirs: $2\beta y - \alpha = B_0$, $2\beta y - 2\beta - \alpha = B_0 - 1/c$, $\alpha y - \beta y^2 = Q(w_0)$ (in their notation, $y = w_0$). Consequently $P(\tau) - \log \varphi_0(e^{-\tau}) \rightarrow 0$ (Section 6 below, following this proposition) is exactly their own Proposition 9.1 applied to this identity, not a separate asymptotic claim about P .*

Sketch. Direct substitution, using $\alpha = a$, $\beta = 1/(2c)$ from their equation (9.4) applied to $a = 3$, $\lambda = 2/3$. $\square$

Series-reverting $Y - \log Y = \tau + r$ to second order, $Y = \tau + \log \tau + r + (\log \tau + r)/\tau + O(\tau^{-2} \log^2 \tau)$, and substituting into P gives

$$P(\tau) = -\frac{(\tau + \log \tau)^2}{2c} + \left(1 + \frac{1}{c} - \frac{r}{c}\right)\tau + \left(\frac{1}{2} - \frac{r}{c}\right)\log \tau + C_P + O\left(\frac{\log^2 \tau}{\tau}\right),$$

$$C_P = -\frac{r^2}{2c} + r + ca + \frac{ca^2}{2} - \frac{\log 2\pi}{2} + \frac{\log c}{2} = -0.9576743183133982760669122497855855 \dots$$

Matching the τ and $\log \tau$ coefficients against Berg and Krüppel's γ , δ_{BK} (their equations following (9.6)) reproduces both exactly, and

$$C_P = \varepsilon \log(2\beta) - \frac{1}{2} \log(2\pi) = \log \frac{(2\beta)^\varepsilon}{\sqrt{2\pi}},$$

so $P(\tau) - \log \varphi_{0,\text{bare}}(e^{-\tau}) \rightarrow C_P$ and, using the prefactor-included normalization, $P(\tau) - \log \varphi_0(e^{-\tau}) \rightarrow 0$ with rate $O(\tau^{-1} \log^2 \tau)$: this is precisely Berg and Krüppel's own Proposition 9.1 (asymptotic, by construction, not exact) applied to the identity of Proposition 17, not an independent asymptotic claim about P itself.

Proof of Corollary 3. By Proposition 16 and 17, $S(\tau) = \log \varphi_0(e^{-\tau}) + H(w_0(\tau)) + O(\tau^{-1} \log^2 \tau)$: the periodic factor multiplying Berg and Krüppel's asymptotic solution, in equation (9.7) for this eigenfunction, is H , matching the product formula their Proposition 9.3 supplies for it, as noted after Theorem 4. Proposition 6's Fourier series is the new, closed form; Proposition 8's certificate is what neither their product formula nor their closing remark (that they only *expect* φ bounded and bounded away from zero relative to φ_0) settles. $\square$

7. PROOFS OF THE MAIN THEOREMS

Proof of Theorem 1. Wirsching's class $\tilde{A}_\delta$ (his equations (1.5) and (3.2)) consists of sequences (x_l) with $\lfloor 3^l x_l \rfloor = k_l$ satisfying $|l - k_l| \leq \delta \sqrt{l}$ for every l ; hence $\lambda_l := 3^l x_l / l \rightarrow 1$ uniformly at rate $O(\delta/\sqrt{l})$ on the class. Writing $z_l = \lambda_l l 3^{-l}$, $\tau_l = -\log z_l = lc - \log l - \log \lambda_l$, the saddle location satisfies $w_0(\tau_l) = lc - \log \lambda_l + O(1/l)$ by Lemma 15's defining equation, so $w_0(\tau_l) \bmod c \rightarrow 0$ uniformly on the class, at rate $O(1/\sqrt{l})$. By Theorem 13, Proposition 16, and continuity of H (indeed Lipschitz, by (3)),

$$\log \varphi(z_l) - \log \varphi_0(z_l) = H(w_0(\tau_l)) + o(1) \rightarrow H(0)$$

uniformly on $\tilde{A}_\delta$, i.e. $\varphi(z_l)/\varphi_0(z_l) \rightarrow e^{H(0)}$ uniformly. This is exactly Conjecture 3, with the constant given explicitly. $\square$

Proof of Theorem 2. As $t = e^{-\tau}$ ranges over $(0, 1/2)$ with no restriction, $w_0(\tau) \bmod c$ ranges over a full period (since Φ_0 is a bijection and $\tau \mapsto \tau \bmod c$ composed with Φ_0^{-1} is onto one period as $\tau \rightarrow \infty$), so by the same combination of results used above, $\log \varphi(t) - \log \varphi_0(t) = H(w_0(\tau)) + o(1)$ takes values arbitrarily close to both $\sup H$ and $\inf H$ infinitely often. By Proposition 8, $\text{osc}(H) > 0$ rigorously, so $\varphi(t)/\varphi_0(t)$ does not converge, and $\limsup / \liminf \geq e^{\text{osc}(H)} \geq 1.0004188$. $\square$

Proposition 18 (Wirsching's actual requirement). *Write x_l^+ for the upper endpoint of Wirsching's own covering interval at level l (playing the same role as this paper's z_l). Wirsching's own use of Conjecture 3, via $\varphi'(x) = \frac{9}{2}\varphi(3x)$ on $(0, \frac{2}{3})$ and his own calculation, his equation (7.13), that $\lim_l (2/3^{l+1})\varphi'_0(x_l)/\varphi_0(x_l) = 2/3$ uniformly on the class, reduces to*

$$\limsup_l \Lambda_l < \frac{3}{2}, \quad \Lambda_l := \frac{\varphi(3x_l^+)/\varphi_0(3x_l^+)}{\varphi(x_l^+)/\varphi_0(x_l^+)}.$$

This holds with a wide margin: $\Lambda_l \rightarrow 1$.

Proof. The phase shift under $t \mapsto 3t$, i.e. $\tau \mapsto \tau - c$, is $w_0(\tau - c) - w_0(\tau) + c = O(1/\tau)$ by Lemma 15's defining equation, independently of the phase of τ . By Proposition 16, $\log \varphi(t) - \log \varphi_0(t) = H(w_0(\tau)) + O(1/\tau)$ uniformly in phase, so

$$\log \Lambda_l = H(w_0(\tau_l - c)) - H(w_0(\tau_l)) + O(1/\tau_l) = O(1/\tau_l) \rightarrow 0,$$

using that H is Lipschitz ((3)) and the phase shift above is $O(1/\tau_l)$; hence $\Lambda_l \rightarrow 1$, well inside $\limsup \Lambda_l < 3/2$. $\square$

8. DISCUSSION

These results settle Conjecture 3 alone. Wirsching's positive-predecessor-density argument reduces to three conjectures; Conjectures 1 and 2, concerning respectively a pointwise-versus-average transfer for the Elka functions and a uniformity-near-a-singularity statement for the limiting operator S_∞ , are untouched here, and the full theorem is not established by this paper.

Wirsching stated the weaker Conjecture 3 instead of the stronger $\varphi \sim \kappa\varphi_0$ he considers first. Theorem 2 shows why: the stronger statement fails by a mechanism with a certified, positive amplitude, not by some correctable technical looseness. The class on which the weaker statement holds is exactly the one phase, among the continuum swept by H , at which the ratio converges. That phase sits close to H 's minimum: $H(0) - \min H = 5.25 \times 10^{-6}$ against a total oscillation of 4.19×10^{-4} . We do not know whether this is coincidental.

The rate is not established beyond $o(1)$. The sharper Edgeworth correction of Remark 14, and a correspondingly sharper form of the envelope and Berg–Krüppel comparisons, would give an explicit polynomial rate throughout; neither theorem needs it.

9. CODE AND DATA AVAILABILITY

Every certified or numerically-checked claim in this paper is backed by a runnable script in the accompanying repository, <https://github.com/faculdade/wirsching-conjecture3-proof>, organized by section with a README per script stating what it verifies and how to run it. This includes the interval-arithmetic certificates behind Proposition 8 and the bounds (3), the constant chain behind Theorem 13, and the identity of Proposition 17.

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